

Yujie Cen

PhD Candidate - Computational Materials Science - TU Wien

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Research Focus

Computational thermal transport and MD simulation in low-dimensional nanomaterials; machine-learning interatomic potentials (MLIP) for large-scale atomistic simulation; phonon dynamics and heat conduction in quasi-1D nanostructures via line group symmetry analysis.

Technical Skills

Category	Tools
MLIP Frameworks	NEP (GPUMD) - MACE - NequIP - DeePMD
DFT / MD Codes	VASP - Quantum ESPRESSO - GPUMD - LAMMPS
Python Ecosystem	PyTorch - ASE - Phonopy - pymatgen - NumPy - SciPy
Thermal Transport	Green-Kubo - GKMA - HNEMA - SHC
HPC & Dev Tools	SLURM - Git - Linux - LaTeX - Docker

Education

PhD in Materials Chemistry - TU Wien, Austria - Jul 2023 - Present (expected late 2026) Supervisor: Prof. Georg K. H. Madsen - Institute of Materials Chemistry (E165), TU Wien Co-supervisor: Dr. Jesus Carrete Montana - Instituto de Nanociencia y Materiales de Aragon, CSIC-Universidad de Zaragoza, Zaragoza, Spain

MS in Theoretical Physics - South China University of Technology - Sept 2020 - Jul 2023

BS in Physics - South China University of Technology - Sept 2016 - Jul 2020

Research Experience

PhD Candidate - TU Wien - Jul 2023 - Present

Symmetry-adapted MLIP-driven thermal transport in quasi-1D systems (*in preparation*) I extend Green-Kubo Modal Analysis (GKMA) and HNEMA by replacing the standard eigenvector basis with line-group projector operators, giving every phonon mode a well-defined irrep label. I trained a NEP force field for (10,0) CNT via a two-round active learning scheme (VASP/PBE-sol-D3 labelling, NEP89 as foundation model), achieving energy RMSE 1.04 meV atom⁻¹ and force RMSE 120 meV Angstrom⁻¹. Production GPUMD trajectories (10-40 ns, 200-400 K) validate that irrep-summed $\kappa = 2538 \text{ W m}^{-1} \text{ K}^{-1}$ is invariant across basis constructions, demonstrating physically interpretable symmetry-resolved thermal conductivity.

Ab-initio heat transport in defect-laden quasi-1D systems (*npj Computational Materials, 2026, first author*) Combined DFT phonon calculations, NEP/GPUMD-accelerated MD, and symmetry-adapted Green-Kubo analysis to quantify how point defects activate symmetry-forbidden inter-channel heat transport in CNTs and WS₂/MoS₂ nanotubes. Demonstrated that MLIP-driven MD reproduces DFT-level thermal conductivity while enabling defect-concentration studies at scales inaccessible to ab initio MD.

pulgon_tools (*CPC, submitted, first author*) Lead developer of an open-source Python package for automated line-group detection, irrep decomposition, character table generation, and symmetry-adapted force constant correction in quasi-1D systems. Production-quality codebase with full test suite, CI/CD, and PyPI release; the equivariant basis construction is directly analogous to the SO(3)-equivariant representations used in MACE and NequIP. github.com/pulgon-project/pulgon_tools

Determining ground states of alloys by symmetry-based classification (*Physical Review Materials*, 2022, first author) Developed a symmetry-based enumeration scheme using DFT (VASP) to identify ground-state configurations in multi-component alloys, reducing computational cost by orders of magnitude compared to exhaustive search.

Research Assistant - South China University of Technology - 2020 - 2023

Applied first-principles methods (VASP) to study magnetic phase transitions in La-Mn perovskite membranes under strain and doping, and investigated intrinsic defects in topological material MnSb₂Te₄.

Publications

First-Author Papers

1. **Cen, Y.**, Wieser, S., Madsen, G. K. H., & Carrete, J. (2026). Ab-initio heat transport in defect-laden quasi-1D systems from a symmetry-adapted perspective. *npj Computational Materials*, 12(1), 19.
2. **Cen, Y.**, Wieser, S., Madsen, G. K. H., & Carrete, J. (2025). Pulgon-tools: A toolkit for analysing and harnessing symmetries in quasi-1D systems. *Computer Physics Communications* (submitted). arXiv:2603.28435.
3. **Cen, Y.**, He, C. C., Qiu, S. B., Zhao, Y. J., & Yang, X. B. (2022). Determining ground states of alloys by a symmetry-based classification. *Physical Review Materials*, 6(5), L050801.

Co-authored Papers

4. Wieser, S., **Cen, Y.**, Madsen, G. K. H., & Carrete, J. (2025). Accelerating first-principles molecular-dynamics thermal conductivity calculations for complex systems. *Journal of Chemical Theory and Computation*.
5. Zhu, C., Yan, X., **Cen, Y.**, He, C. C., Zhao, Y. J., & Yang, X. B. (2025). Accelerating the identification of stable configurations in mixed-anion perovskite materials. *Computational Materials Science*, 248, 113564.
6. Li, D. Z., **Cen, Y.**, Wang, X., & Yang, X. B. (2024). Residual entropy of ice: A study based on transfer matrices. *Physical Review B*, 110(19), 195414.
7. Xiong, J., Peng, Y. H., Lin, J. Y., **Cen, Y.**, Yang, X. B., & Zhao, Y. J. (2023). High Concentration Intrinsic Defects in MnSb₂Te₄. *Materials*, 16(15), 5496.
8. Lin, J. Y., Chen, Z. J., Li, G. L., Zeng, J., **Cen, Y.**, et al. (2022). Theoretical study of magnetic phase transition in La₂-3M₁-3MnO₃ (M = Ca, Sr) membranes through strain and doping. *Physics Letters A*, 432, 128010.

Software

- **Cen, Y.**, Wieser, S., Madsen, G. K. H., & Carrete, J. pulgon_tools: A Python package for line group symmetry analysis of quasi-1D nanostructures. github.com/pulgon-project/pulgon_tools
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Skills & Languages

- **Programming:** Python (advanced), Bash, basic C++
 - **Languages:** Chinese (native), English (professional - academic writing and presentations)
 - **Open Source:** Active contributor to PULGON project; pulgon_tools package maintainer
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References available upon request.